

## **INTERVIEW SCRIPT & THINK-ALOUD PROTOCOL**

### **Study Title:**

**Guided Self-Directed Learning via LLMs and the Role of the Instructor**

### **Principal Investigator / Faculty Advisor:**

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### **Estimated Duration:**

**60 Minutes**

## **INTRODUCTION SCRIPT**

Thank you for participating in this study.

Today, you will evaluate the instructor-facing version of Study Assist, an AI-supported learning platform designed to promote self-regulated learning in introductory computer science courses. The purpose of this session is to understand how instructors interpret and utilize AI-generated student learning analytics and how these insights may support classroom teaching and oversight.

During this session, you will interact with a simulated instructor dashboard populated with sample student data. As you navigate the system, please think aloud by verbalizing your thoughts, expectations, observations, and any confusion you experience. There are no right or wrong answers. We are evaluating the system, not you. This session will be audio and video recorded with your consent. You may skip any question or discontinue participation at any time.

Do you have any questions before we begin?

## **THINK-ALOUD INSTRUCTIONS**

As you complete each task, please continuously describe:

- What you are looking at?
- What you think the information means?

- What you expect to happen next?
- Any confusion or concerns you have?
- Features you find useful or unnecessary?
- How you might use this information in your own teaching?

## **TASKS**

### **Scenario 1: Mid-Semester Evaluation (Week 4)**

**Task 1:** Log into the Instructor Dashboard and review the overall course performance summary.

**Task 2:** Navigate to the Student Tracking panel.

**Task 3:** Identify students who may be at risk of failing and explain what information led you to that conclusion.

**Task 4:** Review student reflection logs and describe how you would use this information to support struggling students.

### **Scenario 2: End-of-Semester Evaluation (Week 10)**

**Task 5:** Review the Topics Learned trend graph and evaluate overall class progression.

**Task 6:** Compare individual student performance trends against class averages.

**Task 7:** Review cumulative rewards, quiz performance, and engagement metrics.

**Task 8:** Inspect system-generated assessment logs and identify any student responses that may require instructor clarification or review.

## **POST-TASK INTERVIEW QUESTIONS**

1. Overall, what were your first impressions of the Study Assist Instructor Dashboard?
2. Think back to your most recent CPS 1231 or introductory programming course. How would this tool have been useful to you?
3. Which dashboard metrics did you find most valuable and why?

4. Were there any metrics or visualizations that you found confusing, unnecessary, or missing?
5. How would having access to these learning analytics influence your lecture planning or classroom activities?
6. How do you envision using AI-generated learning data (e.g., milestone retry alerts, reflection logs, quiz trends) to support student success?
7. What classroom policies or expectations would you establish before introducing Study Assist to students?
8. How would you explain Study Assist to first-year students to ensure they use it as a learning aid rather than an answer-generation tool?
9. What concerns, if any, do you have about relying on AI-generated learning analytics?
10. What additional oversight controls or instructor features would you like to see added to the dashboard?
11. How well do the pacing mechanisms and remediation features support different student populations (novices, experienced programmers, non-CS majors, etc.)?
12. How frequently would you monitor this dashboard during a semester?
13. How useful would the flagged-response review process be in helping you detect potential misunderstandings or AI evaluation errors?
14. What modifications would be necessary before you would feel comfortable using this tool in a live course?
15. Is there anything else you would like to share about your experience using Study Assist?

## **CLOSING SCRIPT**

Thank you for your participation. Your feedback will help improve the design of AI-supported educational technologies and inform future research on instructor-centered learning analytics. If you have any questions after today's session, please contact the research team using the information provided in the consent form.